1) Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

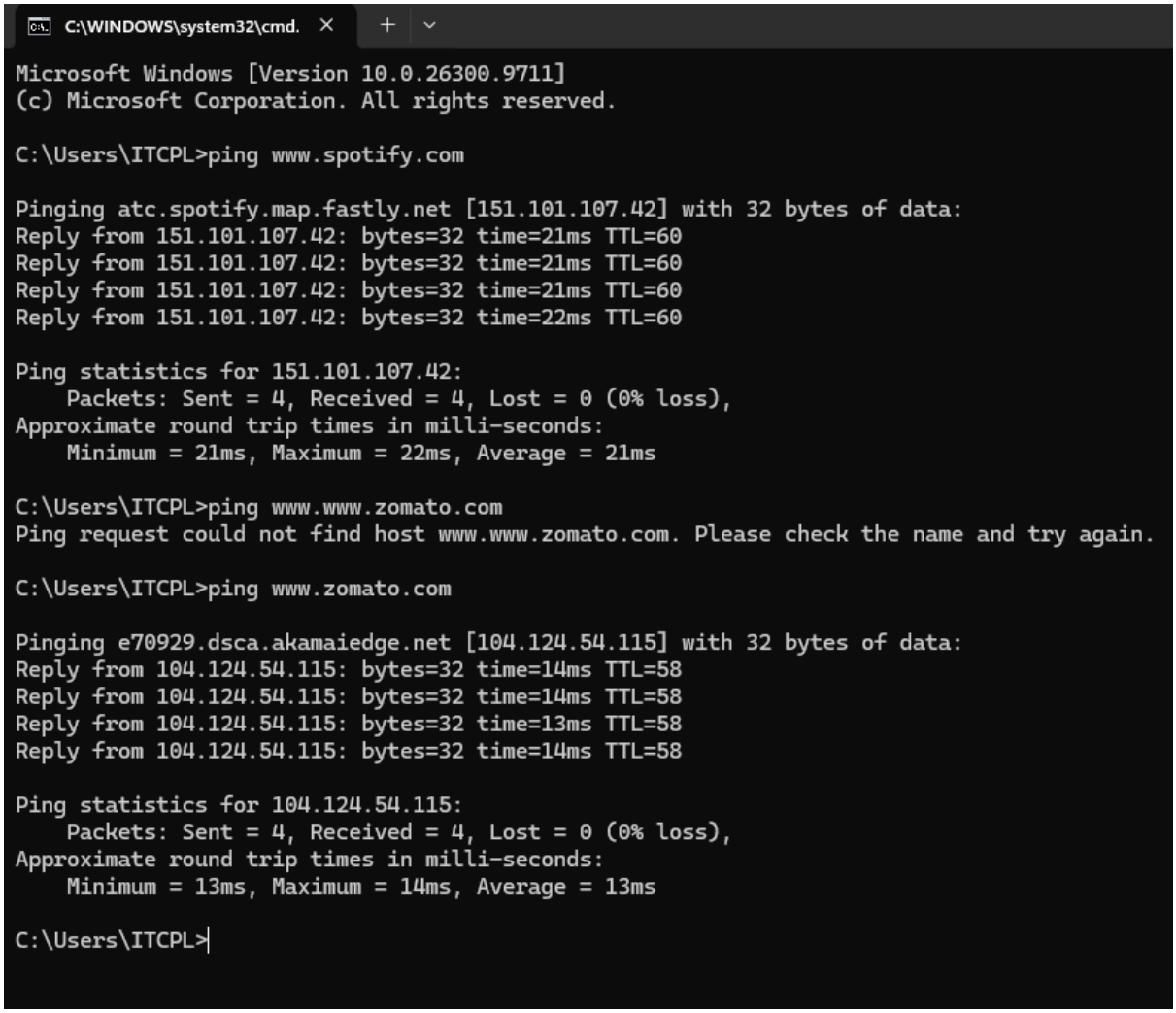


Netwotk traffic .pcapng

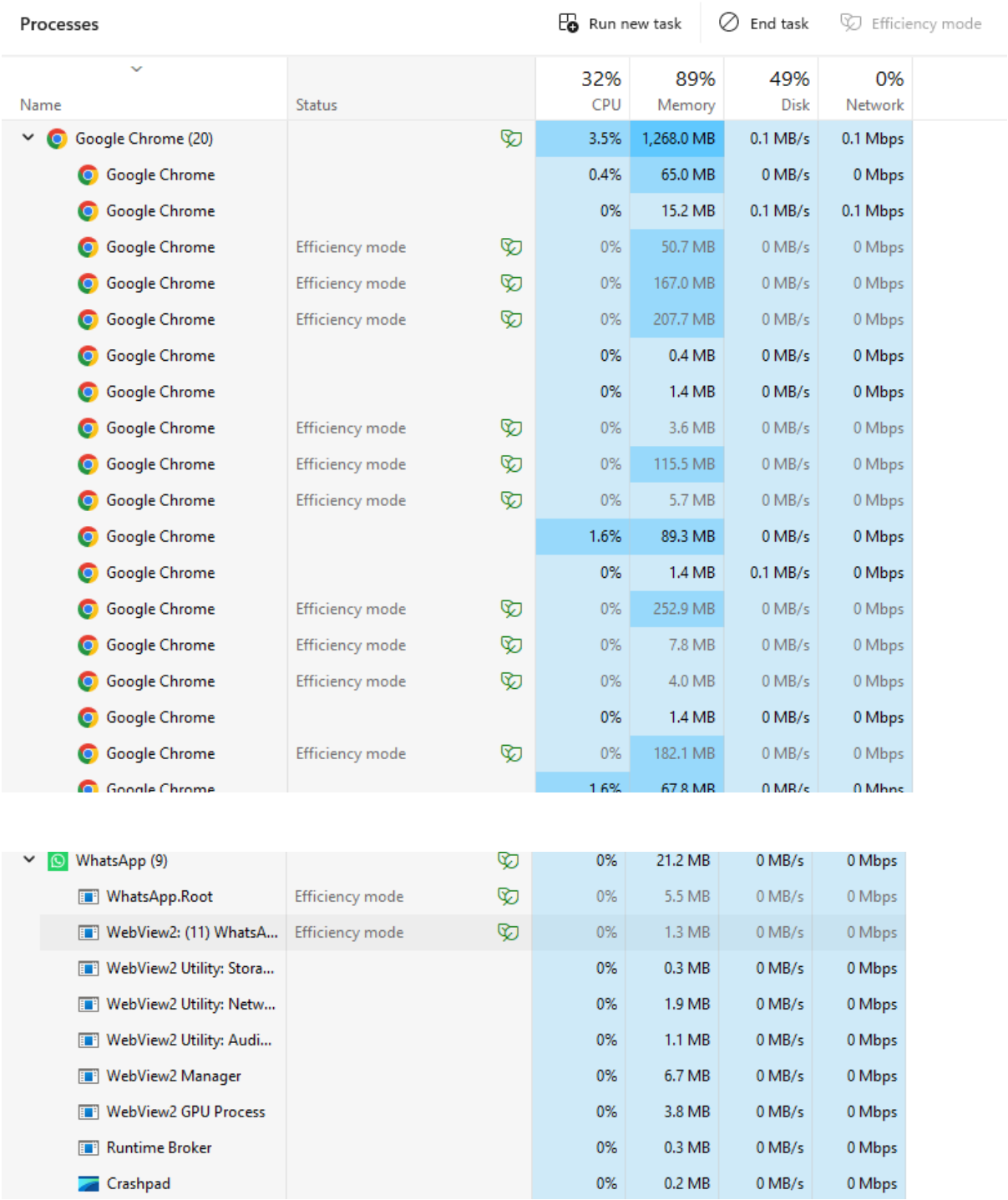
Step 2: Identify the top 3 protocols

Protocol	Purpose
TCP	Provides reliable transport of data between devices.
TLS	Encrypts communication used by HTTPS websites.
QUIC	Supports fast, encrypted connections, commonly over UDP.

2) Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.



3) **Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.**

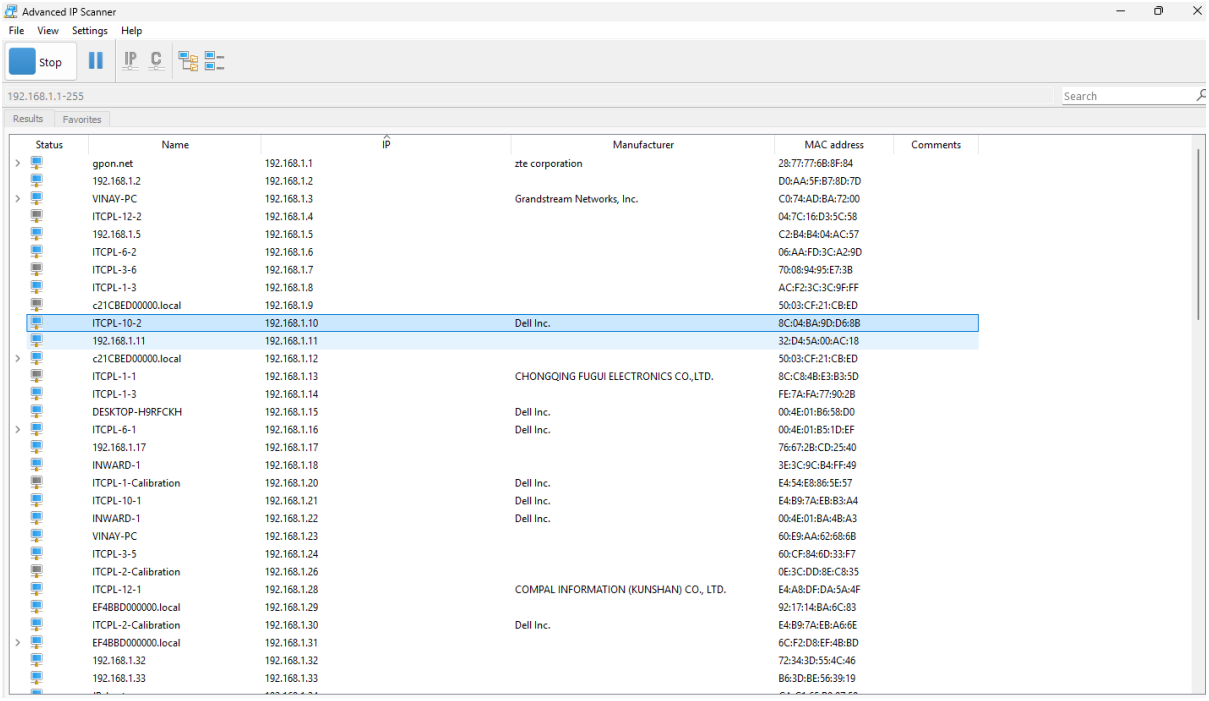


Processes — Run new task — End task — Efficiency mode

Name	Status	32% CPU	89% Memory	49% Disk	0% Network
Google Chrome (20)		3.5%	1,268.0 MB	0.1 MB/s	0.1 Mbps
Google Chrome		0.4%	65.0 MB	0 MB/s	0 Mbps
Google Chrome		0%	15.2 MB	0.1 MB/s	0.1 Mbps
Google Chrome	Efficiency mode	0%	50.7 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency mode	0%	167.0 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency mode	0%	207.7 MB	0 MB/s	0 Mbps
Google Chrome		0%	0.4 MB	0 MB/s	0 Mbps
Google Chrome		0%	1.4 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency mode	0%	3.6 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency mode	0%	115.5 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency mode	0%	5.7 MB	0 MB/s	0 Mbps
Google Chrome		1.6%	89.3 MB	0 MB/s	0 Mbps
Google Chrome		0%	1.4 MB	0.1 MB/s	0 Mbps
Google Chrome	Efficiency mode	0%	252.9 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency mode	0%	7.8 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency mode	0%	4.0 MB	0 MB/s	0 Mbps
Google Chrome		0%	1.4 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency mode	0%	182.1 MB	0 MB/s	0 Mbps
Google Chrome		1.6%	67.8 MB	0 MB/s	0 Mbps
WhatsApp (9)		0%	21.2 MB	0 MB/s	0 Mbps
WhatsApp.Root	Efficiency mode	0%	5.5 MB	0 MB/s	0 Mbps
WebView2: (11) WhatsA...	Efficiency mode	0%	1.3 MB	0 MB/s	0 Mbps
WebView2 Utility: Stora...		0%	0.3 MB	0 MB/s	0 Mbps
WebView2 Utility: Netw...		0%	1.9 MB	0 MB/s	0 Mbps
WebView2 Utility: Audi...		0%	1.1 MB	0 MB/s	0 Mbps
WebView2 Manager		0%	6.7 MB	0 MB/s	0 Mbps
WebView2 GPU Process		0%	3.8 MB	0 MB/s	0 Mbps
Runtime Broker		0%	0.3 MB	0 MB/s	0 Mbps
Crashpad		0%	0.2 MB	0 MB/s	0 Mbps

**4) Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Hint: Make sure your firewall allows scanning, and only scan your own home/college network.**



5) Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

3 Hardware Tools Used in Network Troubleshooting

No.	Hardware Tool	When would you use it?
1	Cable Tester	I would use a cable tester to check Ethernet cables for broken wires, incorrect wiring, or connection faults.
2	Crimping Tool	I would use a crimping tool to attach RJ45 connectors to Ethernet cables when making or repairing network cables.
3	Multimeter	I would use a multimeter to measure voltage, resistance, and continuity when troubleshooting power supply or electrical problems in network equipment.